

Test 2

Instructions

1. As you do the problems on this test, be sure to show your work.
2. If you use your calculator, enter both what you typed into your calculator and the result.
3. If you round, round late and show the value (3 or 4 significant digits) both before and after rounding.

1 Suppose $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cap B) = 0.3$.

- a) What is $P(A \cup B)$?
- b) What is $P(A \mid B)$?
- c) What is $P(B \mid A)$?
- d) Are events A and B independent? Explain how you know.
- e) Are events A and B mutually exclusive? Explain how you know.

2 Let $X \sim \text{Binom}(15, 0.2)$.

- a) What is $P(X = 4)$?
- b) What is $E(X)$?

3 DICE.

- a) If you roll three standard dice (6 sides, each with equal probability), what is the probability that all three numbers will be the same?
- b) If you roll three standard dice, what is the probability that the three numbers rolled will be distinct (no repetitions)?
- c) If you roll three standard dice, what is the probability that at least one of the numbers rolled is a six?
- d) If you roll three standard dice, what is the probability that exactly one of the numbers rolled is a six?

4 The Five Crowns card game uses a non-standard deck. The deck has **five suits** (clubs, diamonds, hearts, spades, and stars). Each suit has **11 kinds** (3 through King, there are no aces and no 2's). There are also jokers, but for this problem they have been removed. That leaves a deck with **55 cards**.

a) If you are dealt six cards, what is the probability that they are all the same suit?

b) If you are dealt six cards, what is the probability that you have two trios? (3 of one kind and 3 of another kind)

c) If you are dealt six cards, what is the probability that you have 3-of-a-kind? (3-of-a-kind means that 3 cards are the same kind, and the other three are different from each other and from the 3 that match.)

Go back and make sure you didn't slip back into standard deck mode anywhere.

5 Calvin has three hats. Each hat contains some maroon and some gold marbles.

- Hat A has 1 maroon and 11 gold.
- Hat B has 1 maroon and 1 gold.
- Hat C has 5 maroon and 1 gold.

Calvin randomly selects a hat (with equal probability) and then randomly selects a marble from that hat.

- a) Altogether, there are 20 marbles. Explain why they are not equally likely to be selected.
- b) What is the probability that the marble selected comes from hat A ?
- c) What is the probability that the selected marble is the maroon marble in hat A ?
- d) What is the probability that the selected marble is maroon (from any hat)?
- e) If the marble selected is maroon, what is the probability that the hat selected is hat A ?
- f) Let M be the event that the marble selected is maroon. Let A be the event that the hat selected is hat A . Circle the correct notation for the probabilities in parts b – e above.

part b: $P(A)$ $P(M)$ $P(A \cap M)$ $P(A \cup M)$ $P(A | M)$ $P(M | A)$

part c: $P(A)$ $P(M)$ $P(A \cap M)$ $P(A \cup M)$ $P(A | M)$ $P(M | A)$

part d: $P(A)$ $P(M)$ $P(A \cap M)$ $P(A \cup M)$ $P(A | M)$ $P(M | A)$

part e: $P(A)$ $P(M)$ $P(A \cap M)$ $P(A \cup M)$ $P(A | M)$ $P(M | A)$

Let W be the amount of money won from one of these lottery tickets.

- ¹This information is available on their website in a different form – and has some errors because they mix up odds and probability. I’ve fixed those errors here.

Space for additional work.

Solutions Problems

1

```
BasicProb
```

```
## # A tibble: 5 x 3
##   probAUB probAgivenB probBgivenA
##   <dbl>     <dbl>     <dbl>
## 1  0.700     0.167     0.500
## 2  0.500     0.500     0.667
## 3  0.500     0.500     0.250
## 4  0.600     0.667     0.400
## 5  0.600     0.750     0.600
```

2

```
dplyr::data_frame(
  n = n,
  p = p,
  k = k,
  prob = dbinom(k, n, p),
  E = n * p
)

## # A tibble: 5 x 5
##       n     p     k prob     E
##   <dbl> <dbl> <dbl> <dbl> <dbl>
## 1  10.0 0.300   5.00 0.103  3.00
## 2  15.0 0.400   6.00 0.207  6.00
## 3  12.0 0.250   4.00 0.194  3.00
## 4  12.0 0.750  10.0 0.232  9.00
## 5  15.0 0.200   4.00 0.188  3.00
```

4

```
choose(5,1) * choose(11,6) / choose(55, 6)

## [1] 0.00007968354

# 3 pairs
choose(11,3) * choose(5,2) * choose(5,2) * choose(5,2) / choose(55, 6)

## [1] 0.005691682

# 2 trios
choose(11,2) * choose(5,3) * choose(5,3) / choose(55, 6)

## [1] 0.0001897227

# 4-of-a-kind
choose(11,1) * choose(5,4) * choose(10,2) * choose(5,1) * choose(5,1) / choose(55, 6)

## [1] 0.002134381

# 3-of-a-kind
choose(11,1) * choose(5,3) * choose(10,3) * choose(5,1) * choose(5,1) * choose(5,1) / choose(55, 6)

## [1] 0.05691682
```

Unused Problems

7 Consider the following game. A player rolls two **fair four-sided** dice and wins the difference between the two values. Here are a few examples:

roll	W
(2, 2)	0
(3, 2)	1
(2, 4)	2

Let the random variable W be the amount won in a single play of the game.

- a) Complete the table below, giving the probabilities of winning various amounts.

Value of W	0	1	2	3
Probability				

- b) What is the expected winnings per play (i.e., the expected value of W)?
- c) If you have to pay \$1 for each roll to play, would you expect to make money or lose money in the long run? How much? (Note: if you pay \$1 to play and then roll (1, 2), you break even.)

8 A fair coin is flipped 6 times.

- a) What is the probability that there are exactly 4 heads?
- b) What is the probability that there are at least 4 heads?
- c) What is the probability that there are at least 4 *consecutive* heads?

9 ABC-123 LOTTERY. Consider the following lottery type game: A player selects (in order) three *distinct* letters from A to Z and three *distinct* digits from 0 to 9. (Examples: one lottery ticket would be AZD-924, another would be ADZ-492. On the other hand, ABA-012 is not a legal ticket because A is repeated and 123-ABC is not a legal ticket because the numbers come first.) Later, the lottery commission draws three distinct letters and three distinct digits at random and in order.

- If all the numbers and letters match in order, then the player wins \$1,000,000.
- If all the numbers and letters match but are not in the correct order, then the player wins \$100,000.
- If the all the letters match (not necessarily in order), but the numbers do not, then the player wins \$1,000;
- If the all the numbers match (not necessarily in order), but the letters do not, then the player wins a prize of \$10;

- a) Fill in the following table:

amount won	1,000,000	100,000	1000	10	0
probability					

- b) Determine the **expected value** of the amount won from one lottery ticket. [This will be graded based on your answers to the questions above, even if they are incorrect, so be sure to show your work.]
- c) How much must the lottery charge per play to avoid losing money at this game?
- d) For which kind of prize should the lottery commission expect to pay out the most money? Explain.

10 Binomial coefficients.

- a) Show how to calculate $C(10, 3) = \binom{10}{3}$ = **without a calculator. Show your work.**
- b) Write a counting problem for which $\binom{10}{3}$ gives the answer. (Keep your problem simple.)

11 This problem is about permutations of the 6-element set $A = \{1, 2, 3, 4, 5, 6\}$

- a) Put the following permutations in lexicographical order: 135246, 124536, 145326, 145362.
- b) List the next **FOUR** permutations in lexicographical order after 124653.

12 Sums.

- a) Express the following sum using $\sum$ notation:

$$5 + 7 + 9 + 11 + 13 + \cdots 5005$$

- b) Express the following sum using $\sum$ -notation:

$$s(n) = 3 + 6 + 12 + \cdots + 3 \cdot 2^n$$

- c) Express the value of $s(n)$ without using $\cdots$ or $\sum$ -notation.

13 A standard deck of cards has 52 cards (4 suits of 13 kinds each). In 5-card poker, two pairs is 2 of one kind of card, 2 of another kind, and 1 of a third kind (so that it doesn't become a full house). (e.g. $5\heartsuit, 5\clubsuit, 8\diamondsuit, 8\spadesuit, 2\spadesuit$) What is the probability of being dealt 2 pairs?

14 Each of these problems deals with probabilities when four 4-sided dice are rolled.

- a) What is the probability that the four numbers rolled will all be the same?
- b) What is the probability that the four numbers rolled will be distinct? (no repetitions)
- c) What is the probability that at least one 1 and at least one 2 are rolled?
- d) Given that at least one 1 and at least one 2 are rolled, what is the probability that all four numbers will be distinct?

15 A bagel shop has 6 types of bagels. How many ways are there to select a ten bagels?